By Nitesh Singh

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SOC

Analyst L1 For Free

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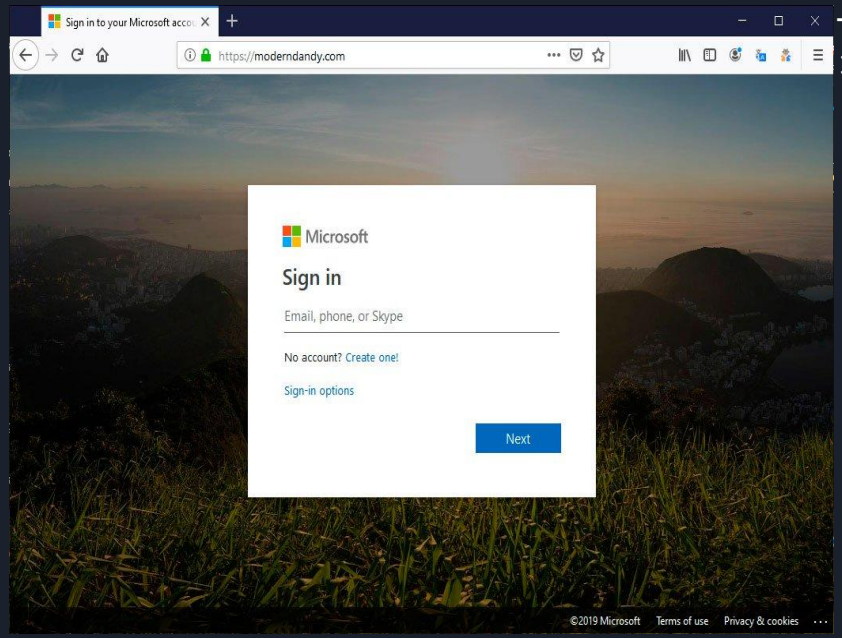
What is Phishing?

Phishing is a **social engineering attack** where attackers trick users into:

- Revealing credentials
- Clicking malicious links
- Downloading malware
- Transferring money
- Installing remote access tools



It attacks **human psychology**, not just systems.

Email Phishing

How It Works:

1. Victim receives fake email
2. Email looks legitimate (bank, Microsoft, courier, etc.)
3. Contains:
    - Malicious link
    - Fake login page
    - Infected attachment
4. User interacts
5. Credentials stolen or malware installed

Common Tactics:

- “Your account will be suspended”
- “Invoice attached”
- “Click to reset password”
- “Salary increment details”
- “Courier delivery failed”

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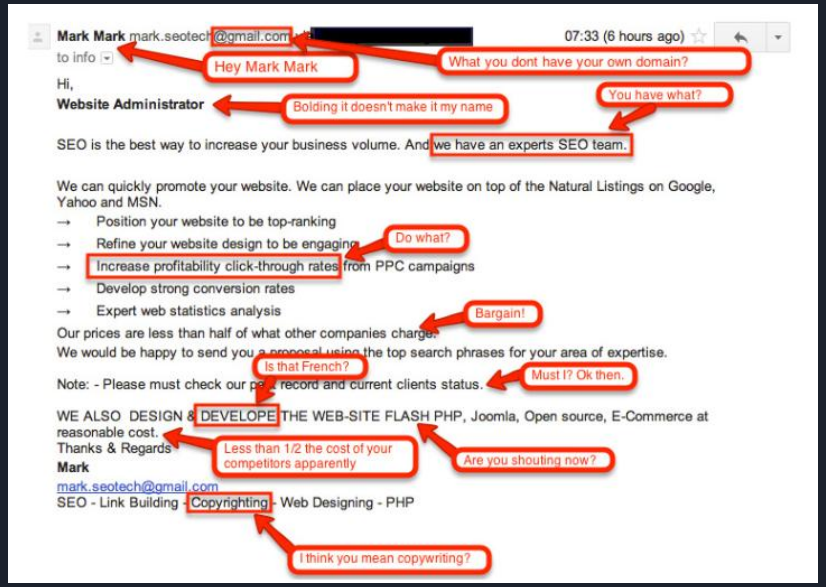
Email Phishing

SOC Detection Indicators:

- Email from spoofed domain
- Domain looks similar (micr0soft.com)
- Suspicious attachments (.zip, .html, .exe, .docm)
- URL redirect chains
- SPF/DKIM/DMARC failure



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Spear Phishing

Definition:

Targeted phishing against a specific person or organization.

Attacker researches:

- LinkedIn
- Company website
- Social media
- Employees

Example:

Attacker emails HR:

> "Hi, this is the CFO. Please transfer ₹5,00,000 urgently."

This is also called **Business Email Compromise (BEC).**

Why It's Dangerous:

- Looks extremely legitimate
- Often bypasses spam filters
- High financial damage

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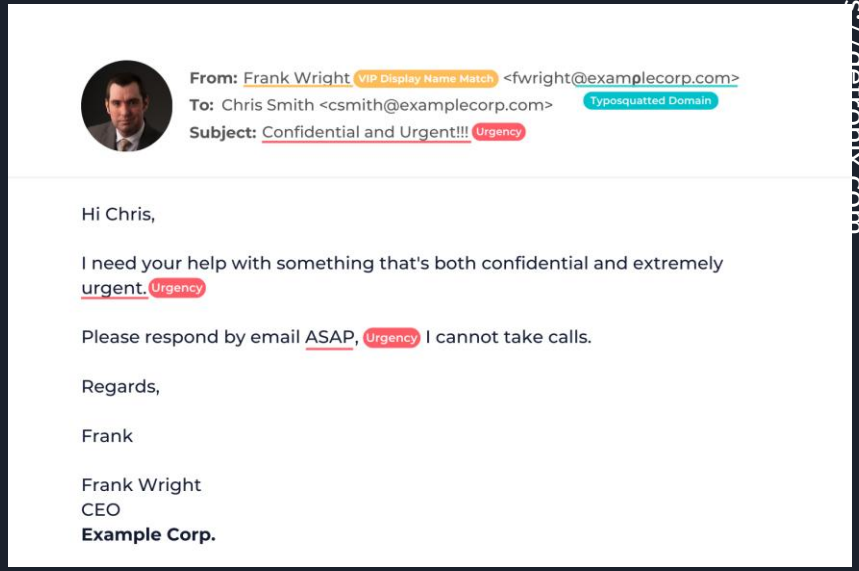
Whaling

Definition:

Phishing targeting **high-level executives** (CEO, CFO, Directors).

High reward, high impact.



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Smishing & Vishing

Smishing:

Phishing via SMS.

Example:

> “Your bank account blocked. Click here.”

Vishing:

Voice phishing via phone call.

Example:

> “I am from bank fraud department. Tell me your OTP.”

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Pharming

What is Pharming?

Pharming is an attack where the victim is redirected to a **fake website** even if they type the correct website address.

Unlike phishing:

- Phishing → User clicks malicious link
- Pharming → Redirection happens silently

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Method 1 - DNS Poisoning

Attacker compromises:

- DNS server OR
- Local router

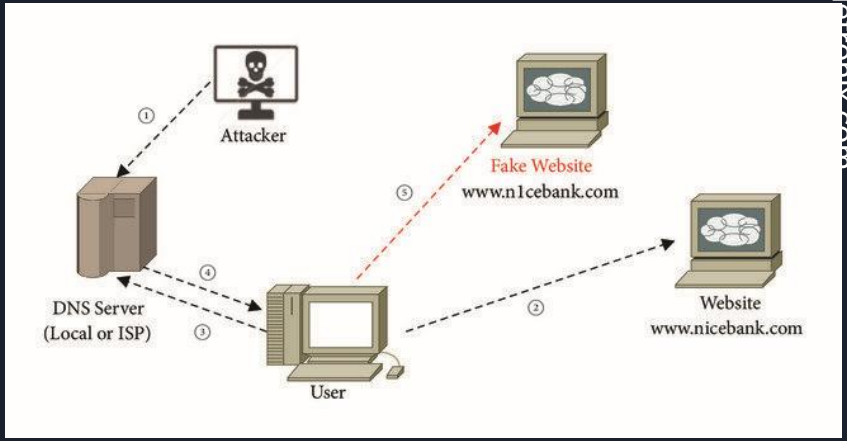
When user types:

`www.bank.com`

DNS returns:

`Attacker IP address`

User lands on fake site.



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Method 2 - Hosts File Modification

Malware edits:

```
C:\Windows\System32\drivers\etc\hosts
```

Example:

```
192.168.1.200    www.bank.com
```

Now user is redirected locally.

Why Pharming is Dangerous

- No suspicious email
- No obvious link
- Looks like real domain
- Harder for users to detect

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SOC Detection Indicators

SOC Detection Indicators

- Sudden DNS record changes
- Suspicious DNS responses
- Internal DNS anomalies
- Multiple users resolving same domain to unusual IP
- Host file modification alerts
- Certificate mismatch errors

Real SOC Scenario

Alert:

Multiple users accessing bank.com resolving to foreign IP.

Investigation:

- Check DNS logs
- Compare with known legitimate IP
- Check for DNS server compromise
- Check endpoint for hosts file tampering

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Evil Twin Attack

What is Evil Twin?

An attacker creates a **fake WiFi access point** that looks identical to a legitimate one.

Example:

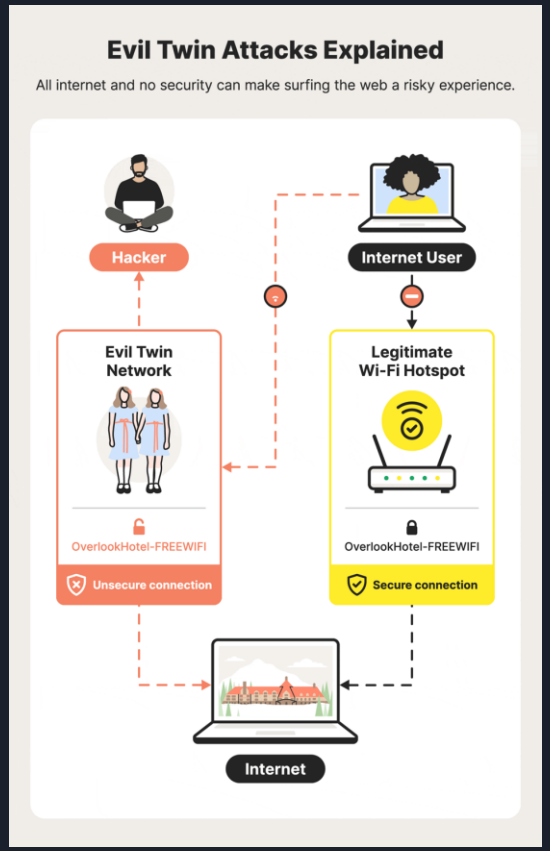
Real WiFi:

`Cafe_WiFi`

Attacker creates:

`Cafe_WiFi_Free`

- Users connect unknowingly.



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How Evil Twin Works

- Attacker sets up rogue access point
- Victim connects
- Attacker intercepts traffic
- Credentials captured
- Session hijacking possible

This often leads to:

- MITM
- Credential theft
- Cookie stealing
- Fake captive portal login pages

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Why Dangerous & Indicators

Why It's Dangerous

- Very easy to execute
- No technical skill required
- Common in:
  - Airports
  - Hotels
  - Cafes
  - Conferences

SOC Detection Indicators

Inside corporate environment:

- Unauthorized access point detected
- Duplicate SSID detected
- ARP spoofing alerts
- Unusual DHCP server detected
- Network traffic interception patterns

Enterprise SOC teams use:

- Wireless IDS (WIDS)
- Network Access Control (NAC)
- Certificate-based WiFi auth (802.1X)

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What is Social Engineering?

Social Engineering = Psychological manipulation to trick people into:

- Revealing sensitive info
- Granting access
- Performing security-breaking actions
- Installing malware
- Transferring money

It exploits **human vulnerabilities**, not technical vulnerabilities.

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Authority

People obey authority figures.

Example:

> “This is IT department. Share your password for system maintenance.”

Why it works:
Humans are conditioned to obey authority.

Urgency

Removes logical thinking.

> “Your account will be deleted in 10 minutes.”

Victim panics → clicks immediately.

Fear

> “Income tax notice attached.”

Fear disables rational analysis.

Core Psychological Attacks Used by Attackers

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Scarcity

“Only 2 bonus slots remaining.”

Triggers impulsive decisions.

Reciprocity

“I helped you last week, can you send me this document?”

Creates obligation.

Familiarity & Trust

Attackers often:

- Impersonate colleagues
- Use internal language
- Reference real projects

This lowers suspicion

Core Psychological Attacks Used by Attackers

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Types of Social Engineering Attacks

Pretexting

Definition:

Attacker creates a fake scenario (pretext) to gain trust.

Example:

> “Hi, I’m from IT. We’re upgrading VPN. Please confirm your credentials.”

This is planned and researched.

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Types of Social Engineering Attacks

Baiting

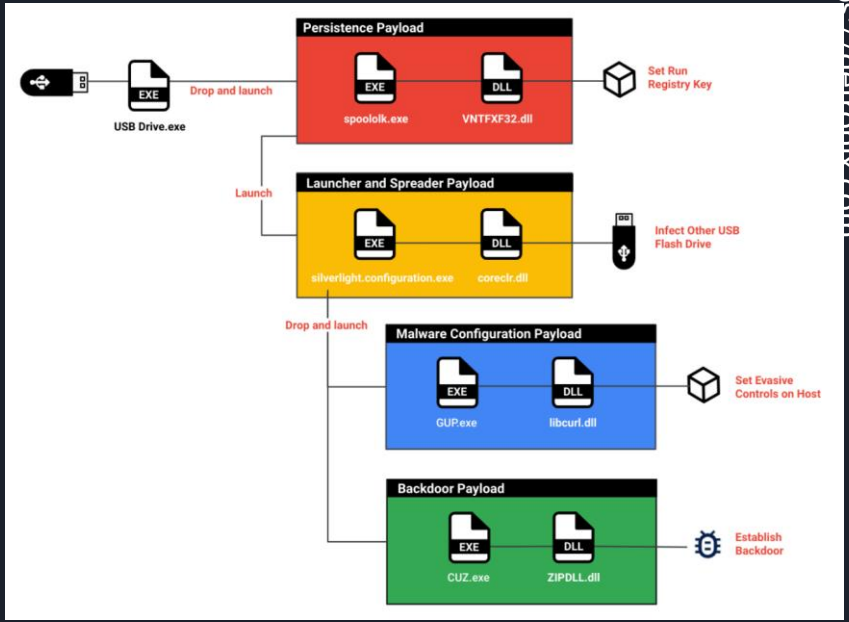
Definition:

Attacker offers something tempting.

Example:

- Free movie download
- Free cracked software
- USB labeled “Salary Data”

Curiosity + greed = compromise.



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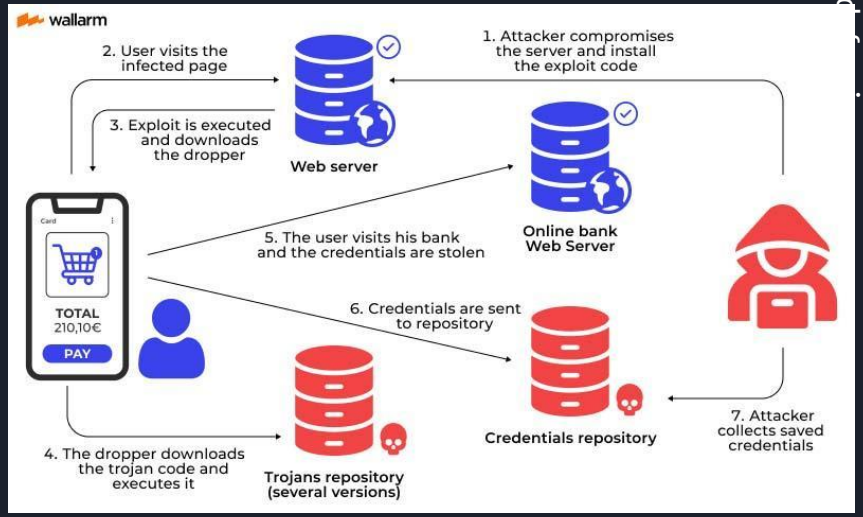
Watering Hole Attack

Attacker compromises a website frequently visited by target group.

Example:

- **Government employees frequently visit specific portal.**
- **Attacker injects malware into that site.**

Very targeted.



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Social Engineering Attack Lifecycle

1. **Reconnaissance**
   - **LinkedIn**
   - **Company website**
   - **Social media**

2. **Relationship Building**
   - **Email conversation**
   - **Phone call**

3. Exploitation
   - Credentials
   - File execution
   - Access granted

4. Exit
   - Delete traces
   - Begin technical attack

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SOC Analyst Perspective

As SOC L1, you usually see:

- Suspicious login after phishing
- VPN login from new location
- Email forwarding rules created
- MFA reset request
- Multiple password reset attempts
- Impossible travel login

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Social Engineering in Indian Context

- Fake GST notices
- Fake income tax emails
- Fake courier delivery scams
- KYC verification fraud
- Bank OTP scams
- Fake job offer scams

Many SOC alerts relate to:

- Credential harvesting
- OTP compromise
- Account takeover

How Organizations Defend

- Security awareness training
- Phishing simulation campaigns
- Email filtering
- MFA enforcement
- Domain monitoring
- Zero trust approach

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